Obsidian suggested practices

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Outline & Goals

Outline

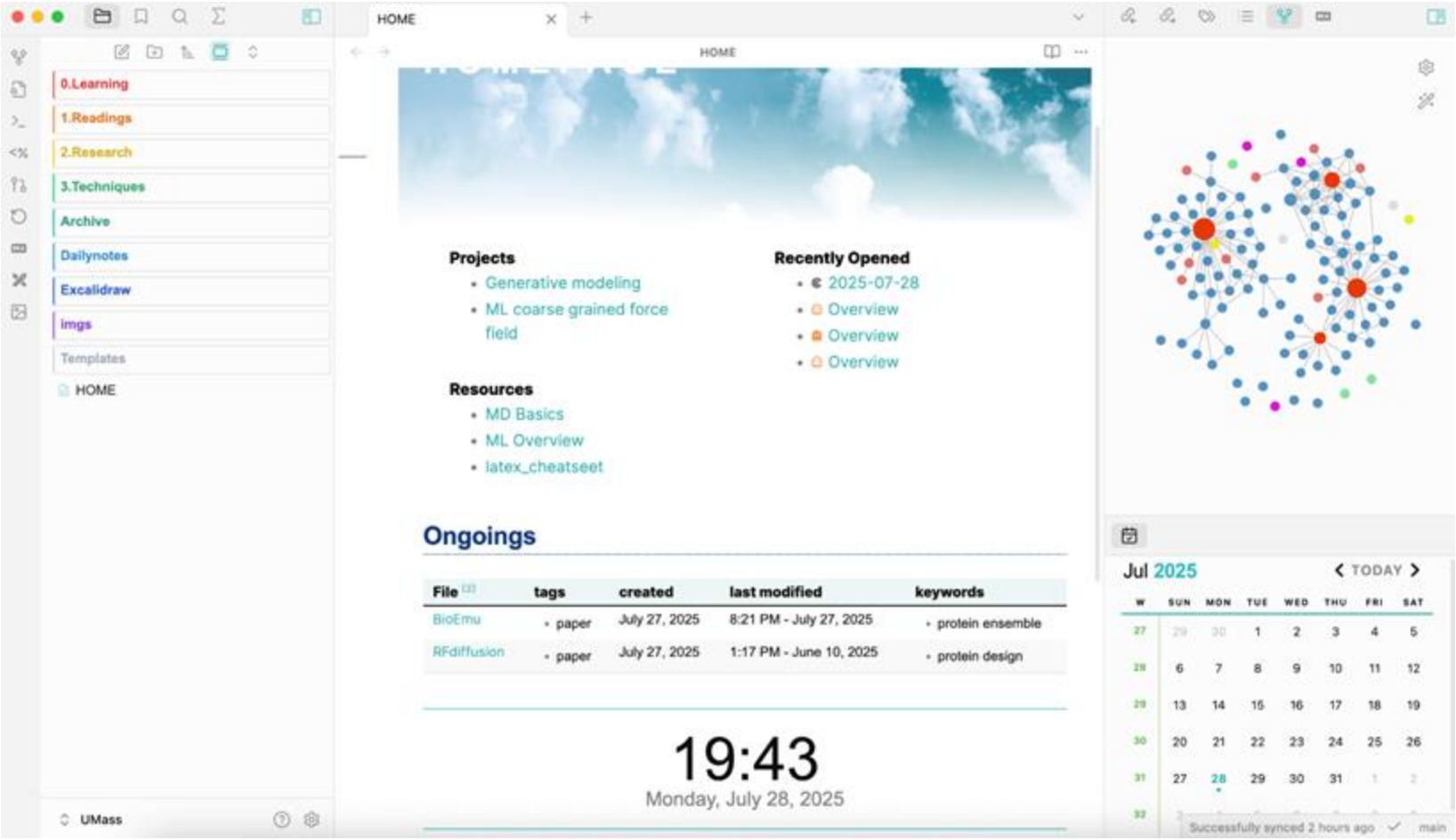
1. Appearance, directory structure and general editor options
2. Link/backline, metadata, tags
3. Synchronization and backup
4. Image management

Goal of this workshop

1. Basic configurations to set up your own obsidian
2. Note organization
3. Plugin suggestions

Introduction

Obsidian is a **markdown**-based, **multi-platform**, **decentralized** and **highly customizable** notebook application



Uniqueness

1. Link/backlink
2. Thousands of plugins
3. Safety and self-owned PI

Not just for taking notes:

1. Databases
2. Goal Management
3. More for you to explore

Appearance and some default behaviors

- **Theme**: controls the overall global appearance

  Settings → Appearance → Themes → manage (recommend: minimal; Blue Topa; Border; Dune etc).

  Explore other options inside Appearance for further fine-tuning

- **Editor**: controls the editing interface appearance

  Settings → Editor, turn on all options except: Right-to-left, Vim key bindings (optional)

    set the Default view for new tabs to be Editing view

    set the Default editing mode to be Live Preview

    set the Properties in document to be Source

  (Other useful options to consider)

  Settings → Files and links

    Default location for new attachments: in the folder specific below (and make folder for this)

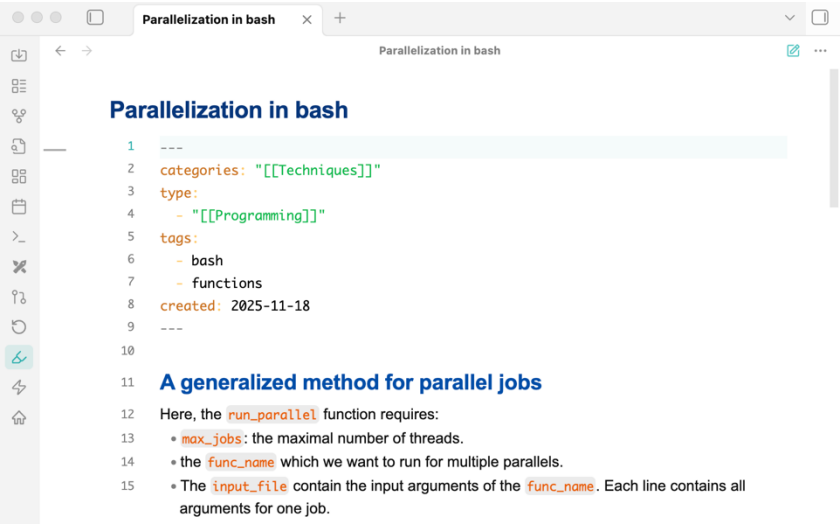
Organization of notes

Basic rules: Discard all directory-based organization and only use Link/backlink and tag in YAML metadata (in the header of a note)

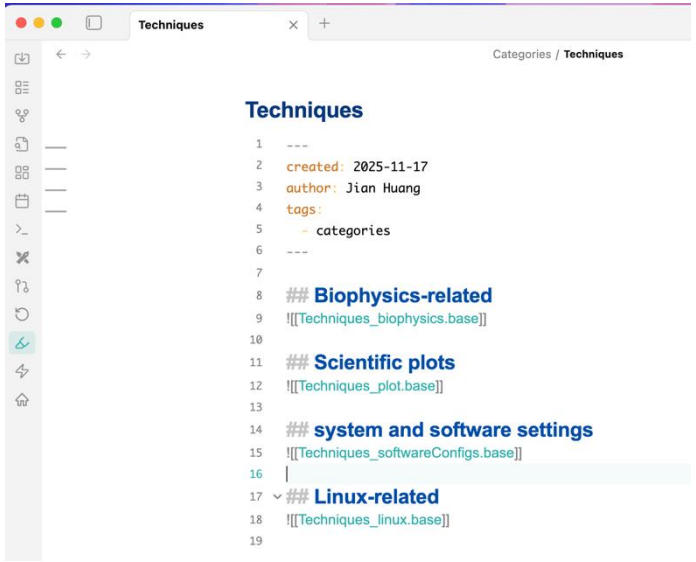
- All note markdown files are under the main vault directory
- Set YAML metadata header for all notes, which includes “tag” entry
- Use **BASE** in obsidian core plugin for querying note lists
- Use Template in obsidian core plugin to quickly generate YAML metadata

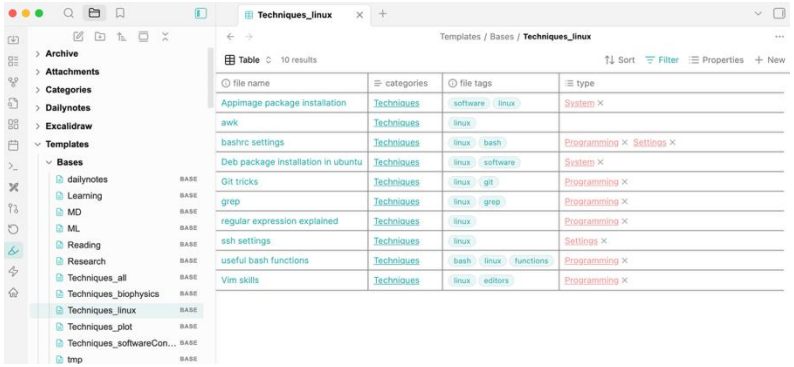
Example:

Metadata



BASE





Link and backlink for referencing

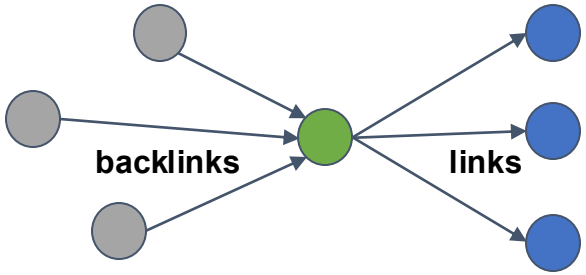
In real life, the “tree” structure might not be sufficient
Use the overview.md for example

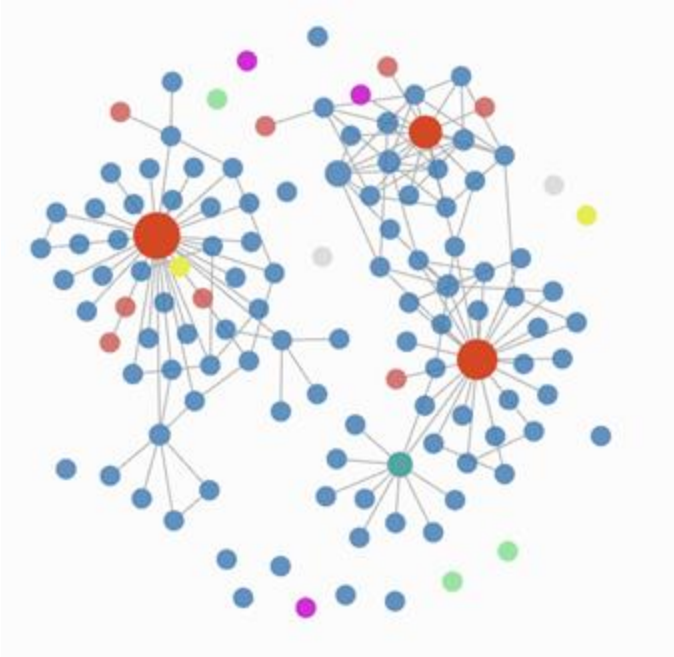
```
# 1. MD simulations
[[MD basics]]
[[Metadynamics]]
[[../0.1 MD simulations/Implicit Solvent Models|Implicit Solvent Models]]
[[../0.1 MD simulations/Parameterization of force field|Parameterization of force field]]
[[../0.1 MD simulations/Steered MD or Umbrella sampling in Gromacs|Steered MD or Umbrella sampling in Gromacs]]

# 2. ML and algorithms
[[../0.2 ML and algorithms/0_HOME/Overview|ML Overview]]

# 3. Music
[[../0.3 Music/music theory|music theory]]

# 4. Economy
[[../0.4 Economy/利率|利率]]

# 5. English
[[../0.6 English/Gathering-01|Gathering-01]]
[[../0.6 English/Gathering-02|Gathering-02]]
```

Source mode (command+shift+E)

Links/backlinks are basically <u>references</u>, allowing for **network**-like structures.
Syntax for creating links/backlinks: **[[note_path]]**

As you can image, a note can link to other notes (“**link**”), while itself can also be linked by other notes (“**backlink**”).





You do not actually need to create a note for using link/backlink

Metadata for your note

Metadata for a note describes the basic information (author, created date, type, status etc).
Using metadata is highly recommended, even more than links/backlinks!

Overview

```yaml
---
created: 2025-07-28
author: Jian Huang
tags: [overview, learning]
project:
type:
status:
---
```

Metadata

```markdown
# 1. MD simulations
[[MD basics]]
[[Metadynamics]]
[[../0.1 MD simulations/Implicit Solvent Models|Implicit Solvent Models]]
[[../0.1 MD simulations/Parameterization of force field|Parameterization of force field]]
[[../0.1 MD simulations/Steered MD or Umbrella sampling in Gromacs|Steered MD or Umbrella sampling in Gromacs]]
```

Actual content

Metadata in Obsidian is defined using the **YAML** format.

Why this is so necessary?

1. can be used with **Dataview** plugin or **BASE** core plugin for sorting and indexing your vault.
2. It is another method for organizing your whole notebook. e.g., tags can be used to cross-referencing and grouping

No worries for typing so many char for the metadata since we have **templater** plugin to automatically generate the code

Synchronization and backup

Basic rules:

1. **Keep three separated copies on** local vaults, cloud vaults and github repositories
2. For daily syncing, always use **Remotely Save to** sync vault to your **cloud folder**
3. Weekly or monthly (or your have major changes in obsidian plugin and settings updates), use git to upload your vault changes to **github repository**

My setup:

1. **ALWAYS** keep a local copy of your vault (could use “git clone” to pull from your github repo.)
   This should be the working environment. You will consistently make changes to your local copy.
1. One copy in my cloud service for syncing across platforms.
   (I am using nutstore; other could services also work, like onedrive)
1. A private remote github repo for version control.
   (Only manually run monthly)

Synchronization and backup

Use cloud service for syncing and github for version control.
Install the **Remotely save** and **Git** plugin in Obsidian.

Remotely Save

Choose A Remote Service

Choose A Remote Service
Start here. What service are you connecting to? S3, Dropbox, Webdav, OneDrive for personal, or Webdis?
Webdav

Remote For Webdav

Disclaimer: The information is stored locally. Other malicious/harmful/faulty plugins may read the info. If you see any unintentional access to your webdav server, please immediately change the username and password.

We will create and sync inside the folder /UMass on your server.

Server Address
Server address.
https://dav.jianguoyun.c

Username
Username. Attention: the username and other info are saved locally.
••••••••••••••••••••••

Password
Password. Attention: the password and other info are saved locally.
••••••••••••••••

Auth Type
If no password is provided, this option will be ignored.
basic

Depth Header Sent To Servers
Webdav servers should be configured to allow requests with header Depth being '1' or 'Infinity'. If you are not sure what's this, choose "depth='1'". If you are sure your server supports depth='infinity', please choose that to get way better performance.
only supports depth='1'

X-Header1: Value1

Set "Schedule For Auto Run" to be 30 min

For Git: mute all other options, except the following:

Push on commit-and-sync
Most of the time you want to push after committing. Turning this off turns a commit-and-sync action into commit and pull only. It will still be called commit-and-sync.

Pull on commit-and-sync
On commit-and-sync, pull commits as well. Turning this off turns a commit-and-sync action into commit and push only.

Source control view

Automatically refresh source control view on file changes
On slower machines this may cause lags. If so, just disable this option.

Show status bar
Obsidian must be restarted for the changes to take affect.

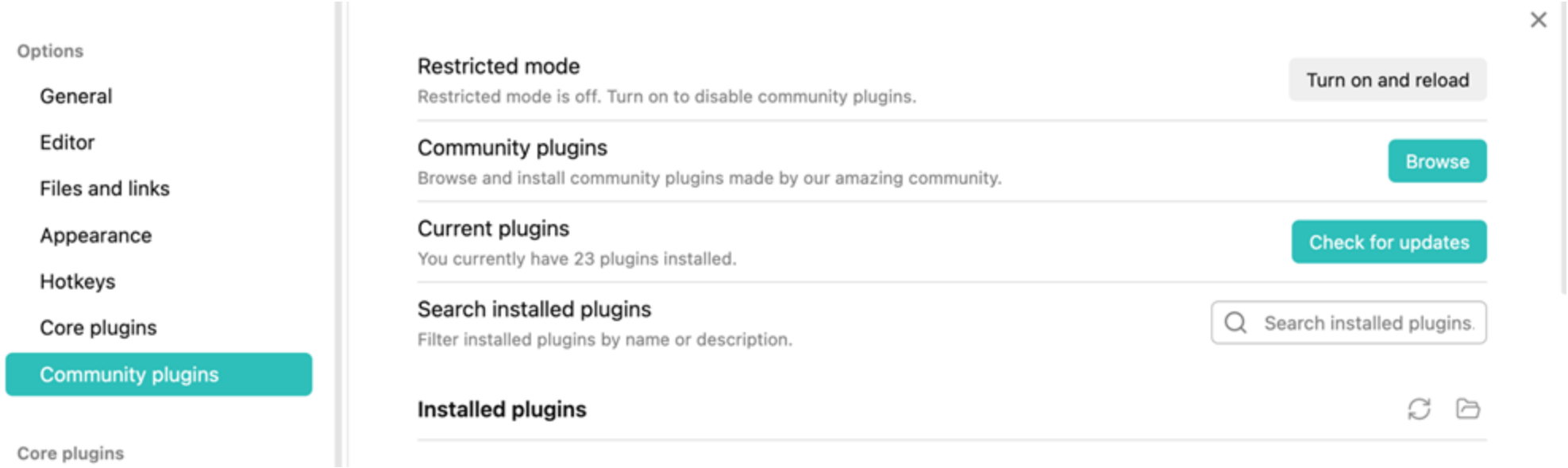
File menu integration
Add "Stage", "Unstage" and "Add to .gitignore" actions to the file menu.

Show branch status bar
Obsidian must be restarted for the changes to take affect.

In principle, run "commit-and-sync" once per month

Use powerful community plugins

Turn on community plugins: go to Settings → Community plugins → turn off Restricted mode
Then we can use the Browse to search for plugins



Use hotkeys to simplify your life

command/ctrl + o: switcher

command/ctrl + p: command palette

customize your own hotkeys: Settings → Hotkeys

Caveat: for Mac users, do NOT use the crtl key (conflicts with windows crtl key)

Image management

Locally: (More recommended!)

use **Image converter** plugin

Caveat: difficult for sharing and publishing your notes to others (but the **share notes** plugin can help)

You can find more details here:

https://share.note.sx/21at8p4b#ejb8TCuj8pzCqSCRp9+mVAzRO70/lKymrUsU3rN1Ofw

Remote: (more steps but convenient for sharing and publishing; less recommended.)

create a github repository for images (has to be a public repo);

download **PicGo** and configure it to connect with your image github repo;

download **the Image auto upload plugin** in Obsidian

Done! [you need to turn off Image converter plugin if you installed and enabled]

Caveat: images are public

Other useful plugins

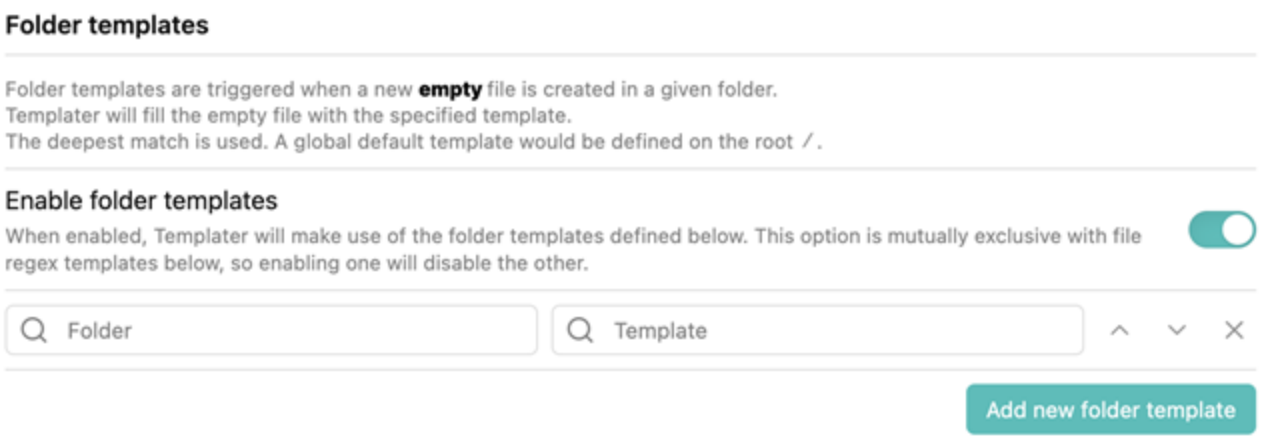
1. **Calendar**

2. **Dataview**
3. **Excalidraw**
4. Markdown formatting assistant
5. **Mathpad: (similar to the desmo tool)**
6. PDF++: (for PDB annotation and viewing and taking notes)
7. **Share note**
8. Style setting
9. **Templater**
10. **Latex Suite**
11. Banner
12. Homepage

To create your homepage find more here:
https://share.note.sx/6mb3ef8u#AOC/14p7CZagkBPIibMh87YxxdeySrMEgNSsWBa+PB8

Templater

Use Templater plugin to format your note, especially automatically generating metadata



Normal notes

```
1  ---
2  created: <% tp.date.now("YYYY-MM-DD") %>
3  author: Jian Huang
4  tags: [note]
5  project:
6  type:
7  status:
8  ---
9
```

Dataview example

```dataview
TABLE tags, created, file.ctime AS "last modified", keywords
from ""
where status = "ongoing"
    and !contains(file.path, "archive/")
    and !contains(file.path, "Templates/")
sort file.ctime desc
```

key:value pairs defined in your metadata

File (2)	tags	created	last modified	keywords
BioEmu	• paper	July 27, 2025	10:29 AM - July 29, 2025	• protein ensemble
RFdiffusion	• paper	July 27, 2025	10:29 AM - July 29, 2025	• protein design

No need to worry about writing the code. Chatgpt can totally handle it.

Thank you.

Outline

Caveat & Solutions for conflicts

Since your obsidian in all platforms will essentially be individual copies, when you have major plugin updates or changes of workspace settings in one PC, you need to **avoid conflicts** between different platforms.

BE VERY **CAREFUL** ABOUT THIS! Below is what I do:

Assuming I am working on Obsidian on my Mac and I updated lots of settings, downloaded lots of plugins and notes, i.e., **A major version update/change**.

1. synchronize your vault through Remotely Save on the Mac
2. Backup your vault using Git to your remote github repository on the Mac
3. Moving to your Windows or Ubuntu, use the git (not the git plugin inside Obsidian) under your vault directory to do the following:

```
git pull    # try pull the remote synchronized version of your vault
# if the above reported some conflicts, try the following
git fetch origin
git reset –hard orgin/main
git clean -fd
```

1. Run Remotely Save on your windows or ubuntu (you might need to disable the protection on **Abort Sync if Modification Above Percentage** in the Remotely Save settings)